Why do suicide rates increase in the spring and decrease in the fall? Insights from Canadian suicide rate and death rate data spanning 2010 to 2020

James C. Wiley

ORCID: https://orcid.org/0000-0002-5049-573X

Corresponding author email: jamescwiley@outlook.com

This version of the article has been accepted for publication, after peer review but is not the Version of Record and does not reflect post-acceptance improvements, or any corrections. The Version of Record is available online at: https://doi.org/10.1007/s42650-026-00110-5. Use of this Accepted Version is subject to the publisher's Accepted Manuscript terms of use https://www.springernature.com/gp/open-research/policies/accepted-manuscript-terms

The published version is freely available from: https://rdcu.be/fabXs

Published-version citation:

Wiley, J. C. (2026). Why do suicide rates increase in the spring and decrease in the fall? Insights from Canadian suicide rate and death rate data spanning 2010 to 2020. *Canadian Studies in Population, 53*, Article 5. https://doi.org/10.1007/s42650-026-00110-5

Accepted: March 17, 2026

Published: March 26, 2026

Why do suicide rates increase in the spring and decrease in the fall? Insights from Canadian suicide rate and death rate data spanning 2010 to 2020

Abstract

We examine the hypothesised population dynamic that non-suicide-related deaths systematically remove potential suicides from the population. If true, suicide rates and death rates should exhibit a consistent negative correlation. In Canada, weekly suicide and death rates display stable seasonal patterns that typically fluctuate in opposite directions. However, the first year of the COVID-19 pandemic disrupted this pattern, with elevated mortality occurring during atypical times of the year. This disruption provides a unique natural experiment to test whether corresponding changes occurred in suicide rates. Using Canadian time-series data from 2010 to 2019, we model weekly suicide rates as a function of death rates, and then apply this model to predict suicide rates for 2020 using observed 2020 death data. Results show that, on average, suicide rates decrease by 0.28 standard deviations per 1 standard deviation increase in the death rate. The model accounts for 23.6% of the variance in observed 2020 suicide rates. Notably, a sharp increase in death rates during the spring of 2020 coincided with a suppression of suicide rates, and this suppression was captured by the model. These findings support the existence of a demographic mechanism in which elevated non-suicide mortality reduces the number of individuals at risk of suicide, thus dampening suicide incidence during periods of high mortality. Further analysis suggests that higher death rates during colder seasons, likely driven by communicable disease transmission and increases in circulatory system diseases, coincide with suppressed suicide rates. This attributes certain fluctuations in suicide rates to competing risks population dynamics.

Keywords: COVID-19, competing risks, death-as-censorship, pandemic, suicide

Résumé

Nous examinons l'hypothèse selon laquelle les décès non liés au suicide soustraient systématiquement de la population des personnes susceptibles de se suicider. Le cas échéant, les taux de suicide et les taux de mortalité devraient présenter une corrélation négative constante. Au Canada, les taux hebdomadaires de suicide et de mortalité présentent des tendances saisonnières stables, fluctuant généralement dans des directions opposées. Néanmoins, au cours de la première année de la pandémie de COVID-19, ce modèle a été perturbé, une mortalité élevée ayant été observée à des périodes inhabituelles de l'année. Cette perturbation représente une expérience naturelle unique permettant de vérifier si des variations connexes se sont produites dans les taux de suicide. Nous avons ainsi modélisé les taux hebdomadaires de suicide en fonction des taux de mortalité à partir de données chronologiques canadiennes de 2010 à 2019, puis utilisé ce modèle pour prédire les taux de suicide de 2020 en nous fondant sur les données de mortalité observées. Les résultats démontrent qu'en moyenne, les taux de suicide diminuent de 0,28 écart-type lorsque le taux de mortalité augmente d'un (1) écart-type. Le modèle explique 23,6 % de la variance des taux de suicide observés en 2020. Une forte augmentation des taux de mortalité au printemps 2020, coïncidant avec une diminution des taux de suicide, a également été décelée. Ces résultats soutiennent l'existence d'un mécanisme démographique selon lequel une augmentation de la mortalité autre que celle liée au suicide diminue le nombre de personnes à risque, réduisant ainsi l'incidence du suicide durant les périodes de forte mortalité. Une analyse plus approfondie suggère que les taux de mortalité plus élevés durant les saisons froides, probablement causés par la transmission de maladies transmissibles et l'augmentation des maladies de l'appareil circulatoire, coïncident avec des taux de suicide plus faibles. Ces résultats

attribuent certaines fluctuations des taux de suicide à une dynamique de population fondée sur les risques concurrents.

1 Introduction

The present paper proposes that non-suicide-related deaths systematically remove individuals from the at-risk population for suicide. That is, every non-suicide-related death could have been a suicide had the individual survived for longer. The death rate is thus a pseudo-measure of censorship (as referred to in mortality analysis; Andersen et al., 2012; Heeringa et al., 2017), wherein early loss of life prevents researchers from knowing if the deceased would have later died by suicide. We will refer to this concept as *death-as-censorship*. We note that death-as-censorship is a recontextualisation of select aspects of competing risks modelling (Andersen et al., 2012; Vaupel et al., 1979). In the present paper, we propose that death-as-censorship can translate into changes in observed suicide rates; this is our central hypothesis, which we evaluate by examining several testable implications. Whether death-as-censorship changes observed suicide rates depends on whether suicide risk positively covaries with other-cause mortality risk (Vaupel et al., 1979); under this overlap, increases in other-cause mortality preferentially remove individuals at higher suicide risk, thereby lowering the expected suicide rate.

A combination of established empirical findings supports this assumption of covarying risks. To illustrate, sub-populations with jointly higher suicide risk and other-cause mortality risk (relative to the general population) include those enduring severe mental illness (Chesney et al., 2014; Hjorthøj et al., 2017; Walker et al., 2015), cancer (Potter et al., 2023; Allemani et al., 2018), homelessness (Hwang, 2000; Hwang et al., 2009; Nielsen et al., 2011; Nilsson et al., 2014), and many other conditions (Alpert et al., 2022; Athey et al., 2024; Crump et al., 2021; Darvishi et al., 2015; Degenhardt et al., 2011; Elser et al., 2023; Giesinger et al., 2020; Gradus et

al., 2010; Mbizvo et al., 2019; Madsen et al., 2018; Mundt et al., 2024; Lange et al., 2024; Larney et al., 2020; Lynch et al., 2020; Roerecke & Rehm, 2013; Wang et al., 2023; Zhong et al., 2021; Zlodre & Fazel, 2012). These findings indicate that, at least across identifiable strata, higher other-cause mortality risk coincides with higher suicide risk. Accordingly, increases in other-cause mortality are expected to intensify the removal of higher-suicide-risk individuals relative to the general population, preferentially shrinking the pool at risk of suicide (a specific application of Vaupel et al., 1979).

Conforming with this expectation, consistent inverse seasonal patterns occur in death rates (Schanzer et al., 2013) and suicide rates (Woo et al., 2012)—suicides increase in the spring and decrease in the fall, whereas deaths exhibit the opposite seasonal pattern. A study from Kaufman and Fitzmaurice (2021) highlights that this inverse correlation between suicide incidence and other-cause mortality occurs across several northern and southern hemisphere countries. Kaufman and Fitzmaurice (2021) attribute this finding to a confounding effect, suggesting underlying seasonal conditions that causally increase suicide also causally decrease circulatory system disease-related mortality, and vice versa.

Alternatively, the present study proposes that death-as-censorship is the actual cause of the negative correlation between suicide rates and death rates. Further supporting our hypothesis, suicide rates during the COVID-19 pandemic decreased or plateaued across numerous countries (when deaths were elevated; World Health Organization, 2023), relative to pre-pandemic suicide rates (Pirkis et al., 2022). Researchers attributed this finding to communities rallying together in difficult times (Pirkis et al., 2022; Sinyor et al., 2022). However, groups with elevated risk of suicide (e.g., Athey et al., 2024; Walker et al., 2015) also had elevated risk of dying when contracting COVID-19 (e.g., Baillargeon et al., 2020; Ge et al., 2021; Leifheit et al., 2021;

Marquez et al., 2021; Nishimi et al., 2022; Siahaan et al., 2021; Venkatesulu et al., 2021). Thus death-as-censorship could also explain some of the variation in suicide rates occurring during the pandemic. Consequently, we aim to parse the death-as-censorship mechanism from the mechanisms proposed by Pirkis et al. (2022) and Kaufman and Fitzmaurice (2021) by examining seasonality occurring within and between years for suicide rates in conjunction with death rates, using weekly data from Canada (Statistics Canada, 2026a, 2026b). Our tests will specifically compare pre-pandemic rate covariation against pandemic-onset rate covariation.

If death-as-censorship operates as expected, pre-pandemic years should express negative covariation between death rates and suicide rates, and this covariation should present similarly during the outset of the COVID-19 pandemic, when upwards disruptions to death rate seasonality occurred (from March to June of 2020; Statistics Canada, 2021). This would suggest there is nothing special about the mechanism generating suicide rates during the COVID-19 pandemic. Rather, there is a special mechanism (COVID-19) generating high death rates, which then disrupts suicide rates through the death-as-censorship mechanism. We thus predict that the correlational patterns observed in pre-pandemic years between suicide rates and death rates will be maintained in the first year of the pandemic, regardless of disruption to the regularly observed seasonality of these variables (Schanzer et al., 2013; Woo et al., 2012).

Our study design isolates the death-as-censorship mechanism by focusing on variation occurring across seasons, rather than multi-year trends. This makes 2020 the most informative year, because it contains the sharpest disruption to the usual seasonal pattern of deaths following COVID-19's onset in Canada (Statistics Canada, 2021; The Canadian Press, 2020). In later pandemic years, mortality increasingly followed typical seasonal patterns, which makes them less diagnostic for separating death-as-censorship from broader pandemic-era trend explanations

(Pirkis et al., 2022) (inferred from Statistics Canada, 2026a, 2026b). Accordingly, we focus on 2020 for inference, while presenting limited post-2020 results only where they inform model evaluation or add understanding to alternative theories.

Just as there is a need to adjust for age before interpreting fluctuations in suicide rates (e.g., Yu & Chen, 2019), the present findings hint that there may be a similar need to adjust for death rates when analysing suicide rates. The effectiveness of societal-level suicide prevention initiatives (e.g., Pirkis et al., 2024) cannot be effectively appraised if external influences remain unaccounted for. The present findings may reconfigure how data driven policies on population-level suicide prevention are evaluated. Ultimately, the present findings suggest many previous interpretations of suicide rate variation (e.g., Burke et al., 2018) may need revisiting. This will be discussed further after results have been presented.

2 Methods

The present paper examines the time series relationship (Hyndman & Athanasopoulos, 2018) between weekly suicide rates and death rates in Canada, with a primary focus on data spanning 2010 to 2020. Raw death and suicide counts were acquired from the Statistics Canada (2026a) website. These were matched with quarterly population estimates (Statistics Canada, 2026b) to derive suicide rates and death rates. Analyses consist of: initial exploration of time series plots presenting standardised suicide rates and death rates broken down by week, by month, and by year; validation of observed seasonality using moving average models; examining the overall time series correlation (Hyndman & Athanasopoulos, 2018; Kutner et al., 2005) between death rates and suicide rates and the correlation for each year of data; building of a random intercept model using 2010-2019 data (predicting suicide rates using death rates, clustered by year); and using said model to predict 2020 suicide rates based on death rates. We

then present limited prediction results for 2021 and 2022 data for model evaluation purposes. All analyses used standardised suicide and death rates to facilitate comparisons. Further details on the data used and analyses employed are provided below.

2.1 Data

2.1.1 Suicide counts and death counts

Both weekly suicide counts and death counts from 2010 to 2022 were acquired from published Statistics Canada data (Statistics Canada, 2026a). Statistics Canada receives these data from the vital statistics offices of provincial and territorial jurisdictions. Suicide counts were subtracted from death counts to remove double counting between these variables. *Death counts* and *death rates* will always exclude suicides whenever referenced throughout the present paper.

Mortality data reflect the most current information available at the time of release and may change slightly as additional records are incorporated. At the time of analysis, data from 2023 onwards were considered "provisional" by Statistics Canada and may be subject to more drastic changes relative to preceding years. Accordingly, we only used mortality data up to 2022. In addition, suicide classification can involve coroner and medical examiner investigation and may vary in timeliness and certainty across jurisdictions. Consequently, suicides are sometimes under-reported, because intent can be difficult to determine and final determinations may be delayed (Navaneelan, 2017).

2.1.2 Quarterly population estimates

Quarterly population estimates were acquired from published Statistics Canada (2026b) data. These estimates are provided for 3-month timeframes, resulting in 4 population estimates per year. Estimates are modelled from Census data (2026c), with the Census being conducted once every 5 years (e.g., 2016, 2021, etc.). The first, second, third, and fourth quarters within the

year will be respectively referred to as Q1, Q2, Q3, and Q4. The quarter boundary dates are January 1, April 1, July 1, and October 1. Each quarterly population estimate is a point-in-time figure as of the boundary date that starts that quarter (e.g., Q3 = July 1; Q4 = October 1).

2.1.3 Rates and standardised rates

Weekly suicide counts and death counts were first matched with their corresponding quarterly population estimate, and then divided by said estimate to calculate respective rates. For example, suicide counts for weeks running through January to March of 2010 were matched with and divided by the Q1 2010 population estimate of 33,809,016 people; suicide counts for weeks running through April to June of 2010 were matched with and divided by the Q2 2010 population estimate of 33,890,461 people; etc. Matching was done in Microsoft Excel. Both rates were standardised (subtracted by their mean and divided by their standard deviation) before being used for analysis (in R; R Core Team, 2023). For simplicity, all mention of rates from this point onwards will refer to the standardised rates unless stated otherwise.

2.2 Analyses

All analyses were conducted in R (R Core Team, 2023), using ggplot2 (Wickham, 2016), dplyr (Wickham et al., 2023), lubridate (Grolemund & Wickham, 2011), lme4 (Bates et al., 2015), Zoo (Zeileis & Grothendieck, 2005), and lmerTest (Kuznetsova et al., 2017) packages. ChatGPT (OpenAI, 2024; versions 4o and 4.5) was used to assist with writing R code, which was then verified and edited by authors. Most reported p-values represent the probability of observing the absolute value of the given statistic assuming its true value is 0 (as is common in 2-tailed null-hypothesis significance testing; Kutner et al., 2005). The only exceptions are p-values reported for R-squared statistics, derived from testing data (2020). For these, a pseudo-correlation was calculated by taking the square root of the variance accounted for, which was

then tested against the null hypothesis of being equal to or less than zero, using a one-tailed test of statistical significance for Pearson correlations (Kutner et al., 2005). We do not perform these tests for negative R-squared values, as any R-squared value at or below zero automatically indicates poor model performance. Where relevant, statistical significance is asserted at the $\alpha = .05$ level.

2.2.1 Exploratory time series plots

A separate plot was constructed for each year from 2010 to 2020, showcasing covariation in suicide and death rates over time. This analysis visually highlights the persistent pre-pandemic seasonality that was disrupted during the outset of the pandemic, and how suicide rates correlate with death rates. This was done using both weekly and month-averaged rates.

Five-week moving average models (Hyndman & Athanasopoulos, 2018; Kutner et al., 2005) using within-year standardisation were built to further validate that said seasonality and disruptions occurred. This approach yielded a grand moving average model for each pre-pandemic rate and a separate 2020 model for each rate. For the pre-pandemic models, confidence and prediction bands were formed by stringing together pointwise intervals from residual variance for each week of the year (1 to 52), using 2010 to 2019 data ($n = 10$ per week; except for week 1 which only had 9 observations, because there were only 51 weeks of published 2010 data, starting January 9th). For this exploratory analysis, pre-pandemic residuals were calculated using weekly differences between each year's moving average model and the grand moving average model. Prediction bands were used to appraise the statistical significance of differences between these non-parametric models, to validate that 2020 suicide rates and death rates diverged from the seasonality observed in pre-pandemic years.

Within-year standardisation and moving averages were used to isolate underlying seasonality. However, this means the common assumption of independence for statistical tests was violated, given moving average models are autocorrelated by design. Under a truly independent binomial test (Mendenhall et al., 2013), we would need 6 of 52 weeks to exceed their prediction bands to indicate a statistically significant difference between pre-2020 and 2020 seasonality. Using a Monte Carlo simulation to adjust for artifactual autocorrelation, we observe that when using five-week moving averages we require 8 of 52 weeks to exceed their prediction bands to meet our threshold for statistical significance. We will therefore report both the binomial test p-values under an independence assumption and the raw number of exceedances relative to the adjusted cutoff of 8 per 52 weeks.

Exploratory analysis suggested traditional time series methods (e.g., ARIMA modelling; Hyndman & Athanasopoulos, 2018; Kutner et al., 2005) would not effectively forecast 2020 suicide rate seasonality. We thus assume that the death-as-censorship mechanism underlies the observed exploratory results. Accordingly, we moved on to test whether the pre-pandemic relationship between death rates and suicide rates could effectively predict 2020 data. We began by validating that correlational patterns inferred from exploratory time series plots were statistically stable. Note that all subsequent analyses used overall standardised rates (i.e., no smoothing and no within-year standardisation) as inputs.

2.2.2 Correlations

Pearson correlations (Kutner et al., 2005) were calculated for the overall time series data spanning 2010 to 2020, as well as for each year independently. This was done to confirm the inference made when examining exploratory time series plots—that suicide rates covary

inversely with death rates. We then moved on to modelling these correlations in pre-pandemic data, and predicting 2020 suicide rates from this model.

2.2.3 Hierarchical linear modelling

Modelling employed the train-test validation approach commonly used in machine learning (Raschka & Mirjalili, 2017), wherein data are partitioned into training and test sets, models are fit on the training set, and predictive capabilities are evaluated on the test set. A random intercept model (Raudenbush & Bryk, 2002) was fit to the 2010-2019 data, predicting suicide rates using death rates, clustering by year. This modelling approach was selected, having observed similar correlational patterns between suicide and death rates across years during exploratory analysis, despite averages for each rate varying slightly across years. With our focus on weekly patterns within years, we treat year-level trends as random effects, to account for their impacts when computing coefficients of interest. Model performance was then evaluated by predicting 2020 suicide rates and comparing to the actual 2020 suicide rates. Variance accounted for by predictions (R-squared; Kutner et al., 2005) was used to evaluate the overall predictive accuracy of the model. The predictive accuracy of the death rate regression coefficient was evaluated separately by adding the mean of residuals to all predicted values, and then re-computing the variance accounted for (R-squared). This counterbalances any off-setting of predictions relative to actual values emerging due to inaccuracy of the model intercept. Pertaining results showcase how closely predictions match the actual data pattern observed between suicide rates and death rates, after having nullified differences attributable to the model intercept. Lastly, we predict 2021 and 2022 weekly suicide rates using respective death rates, to provide additional insights into the performance of our model.

We define our model as such:

$$SR_{ij} = b_{0j} + b_1 DR_{ij} + r_{ij} \quad (1)$$

$$b_{0j} = \gamma_{00} + u_{0j} \quad (2)$$

where SR_{ij} represents the suicide rate for week i in year j, DR_{ij} represents the death rate for week i in year j, b_1 represents the average change in the suicide rate per 1 standard deviation increase in the death rate, r_{ij} represents the level-one residual, b_{0j} represents the yearly intercept, γ_{00} represents the overall (grand mean) intercept, and u_{0j} represents the year-specific deviation from the overall intercept (and can be understood as the level-2 residual).

2.2.4 Robustness to modelling choices

All analyses were re-run using a logit transformation on the outcome variable (suicide rates), because this transformation has some desirable statistical properties when regressing a proportion (which is limited between 0 and 1; Kutner et al., 2005). No material changes in results were realised in this re-run, thus we maintained presentation of results without the logit transformation. For the hierarchical model, we also tested a binomial random intercept model (Kutner et al., 2005; Raudenbush & Bryk, 2002), which similarly showed no material change in results. Accordingly, we opted to maintain our simpler modelling approach, for ease of interpretation. To verify our results against those of Kaufman and Fitzmaurice (2021), we also tested a Poisson random intercept model, using raw suicide counts and raw death counts. Likewise, this final approach did not materially change model interpretation.

3 Results

3.1 Exploratory time series plots

The overall mean and standard deviation for death rates and suicide rates are presented in Table 1. These metrics were used to compute standardised suicide rates and standardised death rates, which were subsequently used in exploratory time series analysis.

Table 1 Mean and standard deviation for raw weekly suicide rates (per 100,000) and raw weekly death rates (per 100,000), Canada, 2010 to 2020

	Suicide rate	Death rate
Mean	0.23	14.01
Standard deviation (*SD*)	0.03	1.18

Note. 2020 data are included in calculations to facilitate model interpretation (later in the paper). Excluding 2020 data does not drastically alter results. Rates are presented unstandardised per 100,000 people at weekly time intervals.

Exploratory time series results for weekly and month-averaged data are displayed respectively in Figures 1 and 2. The weekly data are more granular but harder to interpret, while the month-averaged data showcase clearer data patterns but present less overall information. We will draw the reader's attention to the month-averaged results. 2010 to 2019 death rates tended to decline during spring months, typically accompanied by an increase in suicide rates. This tended to invert in the fall months—with death rates increasing and suicide rates decreasing. In contrast, 2020 death rates peaked slightly late (in April) and the corresponding suicide rate dropped substantially. Furthermore, 2020 results defy the typical U-shaped death rates and inverted U-shaped suicide rates observed in pre-pandemic years.

To illustrate this, we standardised suicide rates and death rates within each year to squash between year variation, and calculated 5-week moving averages for death rates and suicide rates using 2010 to 2019 data (Hyndman & Athanasopoulos, 2018). The grand means of these moving averages across the weeks of the year (numbered 1 to 52) are displayed in Figure 3. We additionally added 95% confidence bands derived from residual variation (Kutner et al., 2005). Residuals were computed from each year's week-specific moving average values, using the grand moving average model to derive predicted values. We then plotted the 5-week moving average for 2020 suicide rates and death rates against their respective pre-pandemic moving average, displayed in Figures 4 and 5. These latter two plots are presented with 95% prediction

bands (Kutner et al., 2005). Figures 3, 4, and 5 showcase that (a) pre-pandemic death rates and suicide rates respectively express U-shaped and inverted U-shaped seasonal patterns and (b) the 2020 data defy these patterns.

We additionally perform a binomial test (Mendenhall et al., 2013) to observe how unlikely the 2020 moving averages are to have been generated by the same seasonal process as the 2010-2019 moving averages. For 2020 suicide rates, 10 observations out of 52 exceed the 2010-2019 prediction band. Here, the probability of observing 10 or more independent exceedances (under $\alpha = .05$) assuming no change in seasonality is $p < .001$. For 2020 death rates, 20 observations out of 52 exceed the 2010-2019 prediction band. Here, the probability of observing 20 or more independent exceedances (under $\alpha = .05$) assuming no change in seasonality is also $p < .001$. Both the 2020 suicide rate and death rate moving average series show more than 8 exceedances, surpassing our autocorrelation-adjusted threshold for statistical significance. For all analyses moving forward, we return to using the overall standardised suicide rates (within-year standardisation and smoothing were only applied for the demonstrations in Figures 3, 4, and 5).

With the insights gained from Figures 4 and 5, we assert that traditional time series forecasting (e.g., ARIMA modelling; Hyndman & Athanasopoulos, 2018; Kutner et al., 2005) using pre-pandemic seasonality would not yield effective predictions of 2020 suicide rates. Accordingly, we test whether the raw time series correlation (Hyndman & Athanasopoulos, 2018) between death rates and suicide rates in pre-pandemic data can predict unseen 2020 data. This test aims to evaluate whether death-as-censorship underlies the generation of suicide rates. We began by validating that the negative correlations inferred from visual examination of time series plots actually exist.

3.2 Correlations

The overall correlation between suicide rates and death rates for years 2010 to 2020 was found to be -0.22 (t = -5.44, p < .001). Separate correlations for each year are listed in Table 2, alongside their respective t- and p-values.

Table 2 Correlations between suicide rates and death rates by year, Canada, 2010 to 2020

Year	Correlation (r)	t-value	p-value
2010	-0.37	-2.82	0.007
2011	-0.29	-2.20	0.032
2012	-0.21	-1.54	0.131
2013	-0.27	-1.97	0.055
2014	-0.29	-2.11	0.040
2015	-0.35	-2.67	0.010
2016	-0.25	-1.81	0.076
2017	-0.20	-1.47	0.147
2018	-0.31	-2.34	0.023
2019	-0.03	-0.24	0.813
2020	-0.55	-4.69	<.001

Next, we explore whether the unusual time series patterns in 2020 death rates can predict the unusual time series patterns in 2020 suicide rates, by mimicking the covariation between these two variables occurring in pre-pandemic years. This is done via a random intercept model.

3.3 Random intercept model

The random intercept model yields an intraclass correlation coefficient of 10.5%, and random effects account for 10.7% of total variance. We thus opt to keep the random intercept component of our model. Results for fixed effects are displayed in Table 3. Results for the regression coefficient of primary interest (death rates) are as follows: b_1 = -0.28; t = -5.74; p < .001. The overall variance accounted for by fixed effects combined with the random intercept is 11.9%. Fixed effects only accounting for a small increase in explained variance is attributable to the lack-of-fit the model has for 2019 data. To illustrate with a few examples: the best fit achieved in training years is in 2010 data, wherein 9.6% of variance is accounted for; the second

worst fit achieved is in 2016 data, wherein 5.9% of variance is accounted for; and the worst fit is realised for 2019 data, wherein -3.8% of variance is accounted for.

Table 3 Fixed effects from the random intercept model, predicting standardised suicide rates using standardised death rates, Canada, 2010 to 2020

Effect	Estimate	Standard Error (SE)	t-value	p-value
Intercept (γ_{00})	0.01	0.11	0.09	.93
Death rate (b_1)	-0.28	0.05	-5.74	< .001

For 2020, the variance accounted for by model predictions is 23.6% (t = 3.96, df = 50, p < .001), which is higher than the variance accounted for in high-performant training years. For 2020, the mean of residuals from prediction is -0.11. After adding this average to all 2020 predictions, the variance accounted for by (adjusted) predictions increases to 25.0% (t = 4.08, df = 50, p < .001). This indicates that the death rate regression coefficient (b_1) has greater predictive power when tested on 2020 data, once having accounted for inaccuracy in the intercept (γ_{00}). Prediction results for 2020 data are plotted in Figure 6.

The respective predicted R-squared and mean-residual adjusted R-squared for 2021 were -77.0% and 4.6% (t = 1.56, df = 50, p = .067), and for 2022 were 25.3% (t = 4.11, df = 50, p < .001) and 30.3% (t = 4.67, df = 50, p < .001). The respective mean residuals from prediction for 2021 and 2022 were -0.69 and -0.22.

We offer an example interpretation of our model using the standard deviations presented in Table 1 and the coefficients presented in Table 3. In Q1 2016, when the Canadian population was estimated to be 35,871,484 people, a decrease of 423 deaths (a 1 standard deviation decrease) from one week to the next would correspond to 3 additional suicides occurring in the latter week. For additional context, in 2016 there were on average 5,033 deaths per week and 85 suicides per week in Canada.

4 Discussion

Overall, the present study found that suicide rates and death rates have a stable negative correlation with each other. However, a few of the years included in the study display a weak and statistically insignificant correlation between these variables. Furthermore, a simple correlational model trained on pre-pandemic data (2010 to 2019) was able to adequately predict suicide rates in 2020, suggesting the process governing the relationship between suicide rates and death rates was maintained in the first year of the pandemic.

Despite the 2020 data not conforming to the seasonality observed in preceding years, the correlational pattern between death rates and suicide rates observed in previous years still permits moderately effective prediction of average suicide rates in 2020. This is asserted given the suicide rate predictions for 2020 testing data follow the time series moving average of actual suicide rates quite closely (inferable by comparing Figure 4 and Figure 6). This suggests some causal link exists between these variables—it appears unlikely that they coincidentally have covarying seasonality when that seasonality is broken by an unusual spike in death rates, yet their covariation remains unchanged. Given suicide cannot be experimentally manipulated, episodes of tight temporal clustering or suppression provide an opportunity to infer precipitants (Wiley, 2020).

The present findings provide moderate evidence to support the existence of death-as-censorship as a mechanism that impacts suicide rates. Furthermore, the present findings re-situate a long-standing public health puzzle—why suicide rates rise in the spring and decline in the fall (Shapiro, 2019; Woo et al., 2012)—as the consequence of competing risks population dynamics (Andersen et al., 2012; Vaupel et al., 1979).

For context, the Canadian summer spans June to September, fall spans September to December, winter spans December to March, and spring spans March to June (National Research Council of Canada, 2024). Using the capital city of Canada (Ottawa) as a reference point, approximate average temperatures for each season during 2024 were as follows: 19 °C in summer; 5 °C in fall; -7 °C in winter; and 9 °C in spring (WeatherSpark, 2025).

Increased death rates in colder months likely correspond to the increased transmission of communicable diseases (Lagacé-Wiens et al., 2021; Schanzer et al., 2013) and prevalence of circulatory system diseases (Auger et al., 2017; Chen et al., 2016; Sheth et al., 1999) occurring during this time. Increased communicable disease transmission is theorised to occur as a consequence of people being more restricted to the indoors and thus being within closer proximity to possible disease carriers (Lofgren et al., 2007), as well as drier air from indoor heating and winter desiccation facilitating disease transmission. Increased prevalence of circulatory system diseases in winter is attributed to an increased need to shovel snow (intense manual labour; Auger et al., 2017), as well as cold weather incentivising sedentariness, substance use, and other behavioural changes (Heart & Stroke, 2025). Similar to COVID-19, those at risk of dying by suicide (e.g., Athey et al., 2024; Walker et al., 2015) are more likely to die from flu complications (Abdel-Rahman, 2021; Nilsson et al., 2021; Ronaldson et al., 2024) and circulatory system diseases (Correll et al., 2017; DeFilippis et al., 2018; Izzy, 2023; Nilsson et al., 1997). This suggests the death-as-censorship mechanism may be responsible for the ordinary seasonality of suicide rates (observed in pre-pandemic years).

To extend interpretation, some groups at higher suicide risk also tend to be older (Yu & Chen, 2019) and have more physical health problems, and suicide risk has been shown to increase across a wide range of medical conditions (Østergaard et al., 2024). Because age is a

predictor of severe COVID-19 outcomes and death (Romero Starke et al., 2021), a mortality disruption like COVID-19 can disproportionately affect these same higher-risk groups, thus also impacting the suicide rate.

As another potential implication, because suicide can be difficult to carry out and often requires planning (Joiner, 2005; Smith & Cukrowicz, 2010; Soper, 2018), suicidal individuals may engage in behaviours that elevate mortality risk rather than making a direct suicide attempt. Some scholars have examined the idea that self-harming behaviour occupies a continuum, ranging from minor risk-taking all the way up to suicidal acts (Durkheim, 1897/2013; Gaffney et al., 2022, 2023; Hagen et al., 2008; Hezel, 1984). Sub-suicidal behaviours high on this continuum may include drug use that teeters on the brink of overdose, driving recklessly while stressed (Becker's position, discussed critically in Ibled, 2025), or taking fewer protective measures against disease transmission. This idea is integrable to the death-as-censorship hypothesis and may help explain the present findings.

The present study only uses Canadian mortality data. Expanding beyond the Canadian context, the study from Kaufman and Fitzmaurice (2021) indicates that the negative temporal correlation between suicide and circulatory system disease mortality replicates across several countries with differing climates (England and Wales, South Korea, United States, Australia, and Brazil). If additional data are available, it would be interesting to examine if disruptions to seasonal mortality during the COVID-19 pandemic occurred in these countries, and if so, whether suicide rates demonstrated an inverse disruption.

Of interest, a paper published in *Nature Climate Change* from Burke et al. (2018) suggests suicide rates increase with increasing temperatures. Burke et al. (2018) interpret their findings as evidence that climate change is a causal input for suicide. Our study finds that suicide

rates are highest in warmer months, when death rates are low. We further demonstrate evidence that suicide rate fluctuations are caused by fluctuations in death rates, and that time-of-year (which can act as a proxy for temperature) is a confounding factor. Lower temperatures seem to be confounded with higher death rates as a consequence of disease transmission mechanisms (Lofgren et al., 2007) and circulatory system disease exacerbation mechanisms (Auger et al., 2017; Heart & Stroke, 2025). While the effects discovered by Burke et al. (2018) could remain robust to the inclusion of death rates in their model, the present study suggests that death rates would be a critical factor to include in such an analysis.

As further examples of findings that may need revisiting, Pirkis et al. (2024) cite several studies that use changes in suicide occurrence to evaluate the effectiveness of population-level interventions. Zalcman and Mann (2007) reported that suicide mortality increased in Alberta following the privatisation of alcohol sales, suggesting a harmful effect of increased alcohol availability. Similarly, Markowitz et al. (2003) found that higher beer taxes in the United States were associated with reductions in youth suicide, supporting taxation as a protective policy measure.

While the findings of Zalcman and Mann (2007) and Markowitz et al. (2003) may hold under further scrutiny, the present analysis highlights the importance of accounting for background mortality trends. If periods of suicide reduction coincide with increases in non-suicide mortality, intervention effects may be overestimated. Conversely, observed increases in suicide might reflect increasing life-expectancy, rather than the effects of policy changes.

We must note that the present study is limited by the fact that it is a natural (rather than a controlled) experiment. The COVID-19 pandemic was marked by lockdowns, altered patterns of social contact, and changes in health care use (Government of Ontario, 2020; Office of the Prime

Minister of Canada, 2020) that may have obscured the usual seasonal drivers of suicide. While our findings challenge existing theories regarding seasonal fluctuations in suicide occurrence, the temperature mechanisms described by Burke et al. (2018), the seasonal mechanisms specified by Kaufman and Fitzmaurice (2021), as well as other theorised seasonal suicide drivers (e.g., increased pollen counts triggering allergies, exacerbating suicide occurrence; Abramowitz et al., 2025; Qin et al., 2013) should continue to receive empirical attention. Given the present findings were observed under highly unusual pandemic conditions, their generalisability to non-pandemic years is uncertain.

A further limitation of our study is the fundamental nonidentifiability in competing risks: we cannot know whether those who died of non-suicidal causes are the same individuals who would otherwise have died by suicide (Tsiatis, 1975). Given the mutual exclusivity of cause-of-death, there is no direct methodological approach to-date that can bypass this limitation.

Another limiting element of our methods is that our inference is anchored to 2020 seasonal disruptions, providing constrained opportunity for replication and reducing the strength of interpretation. With only 2020 seasonality presenting key diagnostic information for parsing various mechanisms, conclusions should be interpreted as suggestive rather than definitive.

An additional constraint of this study is that rates were computed using crude population denominators and therefore do not adjust for changes in the population age structure over the period examined. To the extent that aging affects mortality (and potentially suicide) risk, some of the observed rate variation may reflect shifting composition rather than changes in underlying risk.

Some further favourability towards the death-as-censorship hypothesis is realised when using our model for interpretation. To illustrate, the example of 423 more deaths resulting in 3

fewer suicides on average (in Q1 2016) presents a reduced suicide-to-death ratio of 0.007 (0.007 more suicides occur per 1 fewer deaths). If we assume death-as-censorship to be the explanatory mechanism, this ratio suggests about 7 in every 1,000 deaths censor a suicide. Comparatively, the suicide-to-death ratio observed from 2010 to 2020 is about 0.016, meaning about 16 suicides occur per 1,000 deaths. Given this ratio of deaths attributable to suicide, it is not absurd to think that deaths would subtract from the number of suicides to a similar degree. Additionally, the suicide rate for 2016 was 0.12 per 1,000 people. In comparing to the rate of 7 suicides censored per 1,000 deaths, we see that those who are likely to die from a non-suicidal cause are also more likely to die by suicide (relative to the general population). This supports the idea that a heightened probability of suicide presents with a statistical signal for other-cause mortality (e.g., Favril et al., 2023; Nafilyan et al., 2023). While interesting, these speculations are drawn from crude comparisons, a more authentic approach requires consideration of the varying probabilities of death versus suicide at different stages of life, with consideration of the moving age distribution of the population under study. Perhaps an analysis of the cumulative probability of suicide across the lifespan and how this is impacted by premature death is the more optimal methodological framework for measuring the effect of death-as-censorship on suicide rates.

This highlights a further limitation of the present study: we did not conduct a conventional competing risks analysis (e.g., Fine & Gray, 1999; Kutikov et al., 2012; Xu et al., 2020). Such analyses typically require individual-level time-to-event data with cause-of-death information, and can more directly quantify parameters of interest in the presence of competing risks. Future work could evaluate the death-as-censorship mechanism using these modelling frameworks where the requisite data structure is available. While the present methods allow for a superficial estimation of the probability of a death censoring a suicide, true competing risks

methods would permit greater accuracy in this, as well as providing estimation of the full probability of suicide, after having accounted for competing risks.

A concrete limitation of our modelling strategy was demonstrated when extrapolation beyond 2020 yielded varied predictive power, with poor performance on 2021 data and fairly good performance on 2022 data. The statistical pattern between death rates and suicide rates within 2021 did not materially differ from those found in low-performant pre-pandemic years (e.g., 2012, 2016, 2019). We know from pre-pandemic data that weekly suicide rates are inherently noisy, and that it takes an expansive multi-year series to stabilise the underlying inverted-U seasonal signal in suicide rates. In this sense, the low mean-residual adjusted R-squared (of 4.6%) achieved when predicting 2021 suicide rates is not on its own concerning. However, the large magnitude of the residuals' mean (of -0.69) in 2021 predictions indicates that actual suicide rates in this year are far lower than predictions from our model. Note that suicide rates being exceptionally low while death rates are exceptionally high (as they were in 2021) conforms with the death-as-censorship hypothesis, the divergence seems to mainly reflect limitations of our statistical model. This result hints that there could be a higher order term that should be included in our model, for which we do not have enough variation in pre-pandemic years to adequately capture. Out of curiosity, we tested this using all data from 2010 to 2022 in a simple fixed effects model ($SR_i = b_0 + b_1DR_i + b_2DR_i^2 + e_i$) and achieved decent overall results, with improvements in predicting 2021 suicide rates, but not to the extent that we would deem 2021 performance acceptable. This suggests that there is possibly an unaccounted-for mechanism driving year-over-year changes in suicide rates, beyond the death-as-censorship effect. Although we believe we have successfully isolated death-as-censorship from other mechanisms (subject to the limitations of our study), this does not preclude those other mechanisms from also operating.

We will thus discuss the alternative mechanisms that could explain suicide variation during the COVID-19 pandemic, especially the overall (year-level) decline in suicide rates (Pirkis et al., 2022), which was mainly ignored in our modelling strategy. Declines in suicide rates are common in trying times, e.g., extreme dips in suicide rates occurred during both the First and Second World Wars (Stengel, 1964).

This result is often attributed to increases in social cohesion (Stengel, 1952), which was the primary mechanism used for interpretation by authors documenting this effect during the COVID-19 pandemic (Pirkis et al., 2022; Sinyor et al., 2022). Some of the granular mechanisms specified by Sinyor et al. (2022) include closer day-to-day contact with family and neighbourhood networks during restrictions, which may have eased loneliness and strengthened a sense of belonging; an emphasis on rapid access to evidence-based mental health care and crisis supports; short-term social and economic protections and pre-existing prevention efforts that could have blunted the translation of distress into deaths; adherence to journalistic standards for communicating about suicide in ways that encourage healthy coping rather than normalise harm; and broader initiatives to sustain social cohesion at moments when communities might otherwise fracture. Collectively, these factors offer non-mortality mechanisms by which suicide risk could have fallen during 2020. A common favourable argument for the social cohesion mechanism is that suicide falls during shared celebrations, e.g., after major sporting wins (Encrenaz et al., 2012). The social cohesion hypothesis originates from the work of Émile Durkheim (1897/2013).

Alternatively, we also consider Stengel's (1952, 1964) hypothesis that suicide becomes rarer in times when the value of life within a society is lower, when death is more common. In the modern literature, Stengel's (1952, 1964) hypothesis is situated under the signalling

perspective of suicide, which houses a variety of interrelated studies and ideas (e.g., Hagen et al., 2008; Syme et al., 2016; Wiley, 2020). Stengel's (1952, 1964) hypothesis derives from his theory that suicide attempts are often a means for reviving crumbling interpersonal or romantic relationships. Stengel (1952, 1964) suggests that a suicide attempt is an appeal to the value held by others of the attempter's life. Accordingly, suicide will become rarer in times when the overall value of life within a society is lower, such as in times of war or genocide.

The signalling perspective of suicide (e.g., Hagen et al., 2008) is reliant on costly signalling theory, popularised in economics (Hagen et al., 2008; Spence, 1973; Kennan & Wilson, 1993). Costly signalling theory suggests that if conflicting interests incentivise distrust between cooperating actors, then imposing costs (harm) on oneself conveys credibility. Costly signalling theory saw early application in modelling worker strikes, to understand how workers foregoing their work and monetary earnings incentivises their employers to engage in bargaining (Hagen et al., 2008; Kennan & Wilson, 1993). Within the context of suicidality, Syme et al. (2016) suggest suicide attempts incurring sufficient risk of death are convincing indicators of one's psychological suffering. As an example of this process, Syme et al. (2016) refer to Pospisil's (1958) account of a woman being forced into a prearranged marriage, who attempted suicide several times, resulting in her parents dissolving the marriage arrangement. While not requiring an evolutionary focus (Soper & Shackelford, 2023; e.g., Rosenthal, 1993), many signalling perspective proponents adopt an evolutionary lens, arguing that suicidal behaviour is a fitness gamble that becomes adaptive when facing highly undesirable (low-fitness or fitness-threatening) circumstances (Gaffney et al., 2022; Hagen et al., 2008; Hagen & Rosenström, 2016; Syme et al., 2016; Tan, 2021; Wiley, 2020).

We note that the mechanisms proposed by Pirkis et al. (2022) and Stengel (1952, 1964) are not mutually exclusive, both may contribute to the observed decline in suicide rates during the pandemic. In addition, a desirable property of both theories is that they are not limited by nonidentifiability—those at risk for suicide need not be the same individuals dying as a consequence of an unusual death mechanism. We offer these alternative mechanisms for consideration in interpreting our findings. Ultimately, there are many compounding mechanisms that contribute to the causation of suicide, there is no guaranteed method for pinpointing these mechanisms and untangling their effects.

5 Conclusion

In closing, this study finds stable negative correlations between suicide rates and death rates across most pre-pandemic years in Canada. When these patterns are modelled, they moderately predict suicide rates during the first year of the COVID-19 pandemic, suggesting that the demographic relationship observed before the pandemic persisted into 2020. To explain this phenomenon, we borrow ideas from competing risks modelling (Andersen et al., 2012; Vaupel et al., 1979). Under this perspective, non-suicide-related deaths are understood to remove individuals from the population at risk of suicide, suppressing suicide incidence during periods of elevated overall mortality. This demographic mechanism offers a new lens through which to interpret seasonal and pandemic-related fluctuations in suicide rates, shifting the focus to population-level dynamics. The findings highlight the importance of considering background mortality as a force that shapes the measurement and interpretation of suicide rates. Lastly, alternative interpretations are presented.

Funding

No funding was acquired for this study.

References

Abdel-Rahman, O. (2021). Influenza and pneumonia-attributed deaths among cancer patients in the United States: A population-based study. *Expert Review of Respiratory Medicine, 15*(3), 393–401. https://pubmed.ncbi.nlm.nih.gov/33107375/

Abramowitz, J., Danagoulian, S., & Fleming, O. (2025). Seasonal allergies and mental health: Do small health shocks affect suicidality? *Journal of Health Economics, 104*, 103069. https://doi.org/10.1016/j.jhealeco.2025.103069

Allemani, C., Matsuda, T., Di Carlo, V., et al. (2018). Global surveillance of trends in cancer survival 2000–14 (CONCORD-3): Analysis of individual records for 37.5 million patients diagnosed with one of 18 cancers. *The Lancet, 391*(10125), 1023–1075. https://doi.org/10.1016/S0140-6736(17)33326-3

Alpert, H. R., Hadland, S. E., Hingson, R. W., & Naimi, T. S. (2022). Alcohol consumption and 15 causes of fatal injuries in the United States: A narrative review. *Alcohol Research: Current Reviews, 42*(1), 07. https://doi.org/10.1016/j.amepre.2022.03.025

Andersen, P. K., Geskus, R. B., de Witte, T., & Putter, H. (2012). Competing risks in epidemiology: Possibilities and pitfalls. *International Journal of Epidemiology, 41*(3), 861–870. https://doi.org/10.1093/ije/dyr213

Athey, A., Shaff, J., Kahn, G., Brodie, K., Ryan, T. C., Sawyer, H., DeVinney, A., Nestadt, P. S., & Wilcox, H. C. (2024). Association of substance use with suicide mortality: An updated systematic review and meta-analysis. *Drug and Alcohol Dependence Reports, 14*, 100310. https://doi.org/10.1016/j.dadr.2024.100310

Auger, N., Potter, B. J., Smargiassi, A., Bilodeau-Bertrand, M., Paris, C., & Kosatsky, T. (2017). Association between quantity and duration of snowfall and risk of myocardial infarction. *CMAJ, 189*(6), E235–E242. https://doi.org/10.1503/cmaj.161064

Baillargeon, J., Polychronopoulou, E., Kuo, Y.-F., & Raji, M. A. (2021). The impact of substance use disorder on COVID-19 outcomes. *Psychiatric Services, 72*(5), 578–581. https://doi.org/10.1176/appi.ps.202000534

Bates, D., Maechler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4 (Version 1.1-34) [Computer software]. *Journal of Statistical Software, 67*(1), 1–48. https://doi.org/10.18637/jss.v067.i01

Burke, M., González, F., Baylis, P., Heft-Neal, S., Baysan, C., Basu, S., & Hsiang, S. (2018). Higher temperatures increase suicide rates in the United States and Mexico. *Nature Climate Change, 8*(8), 723–729. https://doi.org/10.1038/s41558-018-0222-x

Chen, H., Wang, J., Li, Q., Yagouti, A., Lavigne, E., Foty, R., Burnett, R. T., Villeneuve, P. J., Cakmak, S., & Copes, R. (2016). Assessment of the effect of cold and hot temperatures on mortality in Ontario, Canada: *A population-based study. CMAJ Open, 4*(1), E48–E58. https://doi.org/10.9778/cmajo.20150111

Chesney, E., Goodwin, G. M., & Fazel, S. (2014). Risks of all-cause and suicide mortality in mental disorders: A meta-review. *World Psychiatry, 13*(2), 153–160. https://doi.org/10.1002/wps.20128

Correll, C. U., Solmi, M., Veronese, N., Bortolato, B., Rosson, S., Santonastaso, P., … Stubbs, B. (2017). Prevalence, incidence and mortality from cardiovascular disease in patients with pooled and specific severe mental illness: A large-scale meta-analysis of 3,211,768

patients and 113,383,368 controls. *World Psychiatry, 16*(2), 163–180. https://doi.org/10.1002/wps.20420

Crump, C., Sundquist, K., Winkleby, M. A., & Sundquist, J. (2021). Comparative risk of suicide by specific substance use disorders: A national cohort study. *Journal of Psychiatric Research, 143*, 89–95. https://doi.org/10.1016/j.jpsychires.2021.10.017

Darvishi, N., Farhadi, M., Haghtalab, T., & Poorolajal, J. (2015). Alcohol-related risk of suicidal ideation, suicide attempt, and completed suicide: A meta-analysis. *PLOS ONE, 10*(5), e0126870. https://doi.org/10.1371/journal.pone.0126870

DeFilippis, E. M., Singh, A., Divakaran, S., Gupta, A., Collins, B. L., Biery, D., Qamar, A., Fatima, A., Ramsis, M., Pipilas, D., Rajabi, R., Eng, M., Hainer, J., Klein, J., Januzzi, J. L., Nasir, K., Di Carli, M. F., Bhatt, D. L., & Blankstein, R. (2018). Cocaine and marijuana use among young adults with myocardial infarction. *Journal of the American College of Cardiology, 71*(22), 2540–2551. https://doi.org/10.1016/j.jacc.2018.02.047

Degenhardt, L., Bucello, C., Mathers, B., Briegleb, C., Ali, H., Hickman, M., & McLaren, J. (2011). Mortality among regular or dependent users of heroin and other opioids: A systematic review and meta-analysis of cohort studies. *Addiction, 106*(1), 32–51. https://doi.org/10.1111/j.1360-0443.2010.03140.x

Durkheim, É. (1897/2013). *Suicide: A study in sociology* (J. A. Spaulding & G. Simpson, Trans.; Reprint ed.). Snowball Publishing.

Elser, H., Gottesman, R. F., Walter, A. E., Coresh, J., Diaz-Arrastia, R., Mosley, T. H., & Schneider, A. L. C. (2023). Head injury and long-term mortality risk in community-dwelling adults. *JAMA Neurology, 80*(3), 260–269. https://doi.org/10.1001/jamaneurol.2022.5024

Encrenaz, G., Contrand, B., Leffondré, K., Queinec, R., Aouba, A., Jougla, E., Miras, A., & Lagarde, E. (2012). Impact of the 1998 Football World Cup on suicide rates in France: Results from the national death registry. *Suicide and Life-Threatening Behavior, 42*(2), 129–135. https://doi.org/10.1111/j.1943-278X.2011.00076.x

Favril, L., Yu, R., Geddes, J. R., & Fazel, S. (2023). *Individual-level risk factors for suicide mortality in the general population: An umbrella review*. The Lancet Public Health, 8(11), e868–e877. https://doi.org/10.1016/S2468-2667(23)00207-4

Fine, J. P., & Gray, R. J. (1999). A proportional hazards model for the subdistribution of a competing risk. *Journal of the American Statistical Association, 94*(446), 496–509. https://doi.org/10.1080/01621459.1999.10474144

Gaffney, M. R., Adams, K. H., Syme, K. L., & Hagen, E. H. (2022). Depression and suicidality as evolved credible signals of need in social conflicts. *Evolution and Human Behavior, 43*(3), 242–256. https://doi.org/10.1016/j.evolhumbehav.2022.02.004

Gaffney, M. R., Adams, K. H., Syme, K. L., & Hagen, E. H. (2023). Response to: "Are depression and suicidality evolved signals? Evidently, no". *Evolution and Human Behavior, 44*(2), 155–160. https://doi.org/10.1016/j.evolhumbehav.2023.02.003

Ge, E., Li, Y., Wu, S., Candido, E., & Wei, X. (2021). Association of pre-existing comorbidities with mortality and disease severity among 167,500 individuals with COVID-19 in Canada: A population-based cohort study. *PLOS ONE, 16*(10), e0258154. https://doi.org/10.1371/journal.pone.0258154

Giesinger, I., Li, J., Takemoto, E., Cone, J. E., Farfel, M. R., Brackbill, R. M., & Kahn, A. R. (2020). Association between posttraumatic stress disorder and mortality among

responders and civilians following the September 11, 2001 disaster. *JAMA Network Open, 3*(2), e1920476. https://doi.org/10.1001/jamanetworkopen.2019.20476

Government of Ontario. (2020). *Ontario extends emergency declaration to stop the spread of COVID-19*. Ontario Newsroom. https://news.ontario.ca/en/release/56523/ontarioextends-emergency-declaration-to-stop-the-spread-of-covid-19

Gradus, J. L., Qin, P., Lincoln, A. K., et al. (2010). Posttraumatic stress disorder and completed suicide. *American Journal of Epidemiology, 171*(6), 721–727. https://doi.org/10.1093/aje/kwp456

Grolemund, G., & Wickham, H. (2011). lubridate: Make dealing with dates a little easier (Version 1.9.3) [Computer software]. https://lubridate.tidyverse.org

Hassan, L., Sawyer, C., Peek, N., Lovell, K., Carvalho, A. F., Solmi, M., Tilston, G., Sperrin, M., & Firth, J. (2023). Heightened COVID-19 mortality in people with severe mental illness persists after vaccination: A cohort study of Greater Manchester residents. *Schizophrenia Bulletin, 49*(2), 275–284. https://doi.org/10.1093/schbul/sbac118

Hagen, E. H., & Rosenström, T. (2016). Explaining the sex difference in depression with a unified bargaining model of anger and depression. *Evolution, Medicine, and Public Health, 2016*(1), 117–132. https://doi.org/10.1093/emph/eow006

Hagen, E. H., Watson, P. J., & Hammerstein, P. (2008). Gestures of despair and hope: A view on deliberate self-harm from economics and evolutionary biology. *Biological Theory, 3*(2), 123–138. https://doi.org/10.1162/biot.2008.3.2.123

Heart & Stroke. (2025). Don't fall victim to the holiday heart attack. Retrieved October 21, 2025, from https://www.heartandstroke.ca/articles/dont-fall-victim-to-the-holiday-heart-attack

Heeringa, S. G., West, B. T., & Berglund, P. A. (2017). *Applied survey data analysis* (2nd ed.). Chapman and Hall/CRC.

Hezel, F. X. (1984). Cultural patterns in Trukese suicide. Ethnology, 23(3), 193–206. https://doi.org/10.2307/3773746

Hjorthøj, C., Stürup, A. E., McGrath, J. J., & Nordentoft, M. (2017). Years of potential life lost and life expectancy in schizophrenia: A systematic review and meta-analysis. *The Lancet Psychiatry, 4*(4), 295–301. https://doi.org/10.1016/S2215-0366(17)30078-0

Hwang, S. W. (2000). Mortality among men using homeless shelters in Toronto, Ontario. *JAMA, 283*(16), 2152–2157. https://doi.org/10.1001/jama.283.16.2152

Hwang, S. W., Wilkins, R., Tjepkema, M., O'Campo, P. J., & Dunn, J. R. (2009). Mortality among residents of shelters, rooming houses, and hotels in Canada: 11-year follow-up study. *BMJ, 339*, b4036. https://doi.org/10.1136/bmj.b4036

Hyndman, R. J., & Athanasopoulos, G. (2018). *Forecasting: Principles and practice* (2nd ed.). OTexts. https://otexts.com/fpp2/index.html

Ibled, C. (2025). "Most (if not all!) deaths are to some extent suicides": Human capital and endogenous mortality in Gary Becker's work. *Economy and Society*. https://doi.org/10.1080/03085147.2025.2571323

Izzy, S., Grashow, R., Radmanesh, F., Chen, P., Taylor, H., Formisano, R., Wilson, F., Wasfy, M., Baggish, A., & Zafonte, R. (2023). Long-term risk of cardiovascular disease after traumatic brain injury: Screening and prevention. *The Lancet Neurology, 22*(10), 959–970. https://doi.org/10.1016/S1474-4422(23)00241-7

Joiner, T. E. (2005). *Why people die by suicide*. Harvard University Press

Kaufman, M. J., & Fitzmaurice, G. M. (2021). Temporal patterns of suicide and circulatory system disease-related mortality are inversely correlated in several countries. *BMC Psychiatry, 21*, Article 153. https://doi.org/10.1186/s12888-021-03159-5

Kennan, J., & Wilson, R. (1993). Bargaining with private information. *Journal of Economic Literature, 31*(1), 45–104. https://www.jstor.org/stable/2728150

Kutikov, A., Cooperberg, M. R., Paciorek, A. T., Uzzo, R. G., Carroll, P. R., & Boorjian, S. A. (2012). Evaluating prostate cancer mortality and competing risks of death in patients with localized prostate cancer using a comprehensive nomogram. *Prostate Cancer and Prostatic Diseases, 15*(4), 374–379. https://doi.org/10.1038/pcan.2012.21

Kutner, M. H., Nachtsheim, C. J., Neter, J., & Li, W. (2005). *Applied linear statistical models* (5th ed.). McGraw-Hill Irwin.

Kuznetsova, A., Brockhoff, P. B., & Christensen, R. H. B. (2017). lmerTest package: Tests in linear mixed effects models (Version 3.1-3) [Computer software]. *Journal of Statistical Software, 82*(13), 1–26. https://doi.org/10.18637/jss.v082.i13

Lagacé-Wiens, P., Bullard, J., Cole, R., & Van Caeseele, P. (2021). Seasonality of coronaviruses and other respiratory viruses in Canada: Implications for COVID-19. *Canada Communicable Disease Report, 47*(3), 132–138. https://doi.org/10.14745/ccdr.v47i03a02

Lange, S., Kim, K. V., Lasserre, A. M., Orpana, H., Bagge, C., Roerecke, M., & Rehm, J. (2024). Sex-specific association of alcohol use disorder with suicide mortality: A systematic review and meta-analysis. *JAMA Network Open, 7*(3), e241941. https://doi.org/10.1001/jamanetworkopen.2024.1941

Larney, S., Tran, L. T., Leung, J., & Degenhardt, L. (2020). All-cause and cause-specific mortality among people using extramedical opioids: A systematic review and meta-

analysis. *JAMA Psychiatry, 77*(12), 1202–1212. https://doi.org/10.1001/jamapsychiatry.2019.4170

Leifheit, K. M., Chaisson, L. H., Medina, J. A., et al. (2021). Elevated mortality among people experiencing homelessness with COVID-19. *Open Forum Infectious Diseases, 8*(7), ofab301. https://doi.org/10.1093/ofid/ofab301

Lofgren, E., Fefferman, N. H., Naumov, Y. N., Gorski, J., & Naumova, E. N. (2007). Influenza seasonality: Underlying causes and modeling theories. *Journal of Virology, 81*(11), 5429–5436. https://doi.org/10.1128/jvi.01680-06

Lynch, F. L., Peterson, E. L., Lu, C. Y., Hu, Y., Rossom, R. C., Waitzfelder, B. E., Owen-Smith, A., Hubley, S., Prabhakar, D., Williams, L. K., Beck, A., Simon, G. E., & Ahmedani, B. K. (2020). Substance use disorders and risk of suicide in a general US population: A case-control study. *Addiction Science & Clinical Practice, 15*, 14. https://doi.org/10.1186/s13722-020-0181-1

Madsen, T., Erlangsen, A., Orlovska, S., Mofaddy, R., Nordentoft, M., & Benros, M. E. (2018). Association between traumatic brain injury and risk of suicide. *JAMA, 320*(6), 580–588. https://doi.org/10.1001/jama.2018.10211

Markowitz, S., Chatterji, P., & Kaestner, R. (2003). Estimating the impact of alcohol policies on youth suicides. *Journal of Mental Health Policy and Economics, 6*(1), 37–46. https://europepmc.org/article/med/14578546

Marquez, N., Ward, J. A., Parish, K., Saloner, B., & Dolovich, S. (2021). COVID-19 incidence and mortality in U.S. federal and state prisons vs the U.S. population. *JAMA, 326*(18), 1865–1867. https://doi.org/10.1001/jama.2021.17575

Mbizvo, G. K., Bennett, K., Simpson, C. R., Duncan, S. E., & Chin, R. F. M. (2019). Epilepsy-related and other causes of mortality in people with epilepsy: A systematic review of systematic reviews. *Epilepsy Research, 157*, 106192. https://doi.org/10.1016/j.eplepsyres.2019.106192

Mendenhall, W., Beaver, R. J., Beaver, B. M., & Ahmed, S. E. (2013). *Introduction to probability and statistics*. Nelson Education.

Mundt, A. P., Cifuentes-Gramajo, P. A., Baranyi, G., & Fazel, S. (2024). Worldwide incidence of suicides in prison: A systematic review and meta-analysis. *The Lancet Psychiatry, 11*(7), 536–544. https://doi.org/10.1016/S2215-0366(24)00134-2

Nafilyan, V., Morgan, J., Mais, D., Sleeman, K. E., Butt, A., Ward, I., Tucker, J., Appleby, L., & Glickman, M. (2023). Risk of suicide after diagnosis of severe physical health conditions: A retrospective cohort study of 47 million people. *The Lancet Regional Health – Europe, 25*, Article 100562. https://doi.org/10.1016/j.lanepe.2022.100562

National Research Council of Canada. (2024). *When do the seasons start?* Government of Canada. https://nrc.canada.ca/en/certifications-evaluations-standards/canadas-official-time/3-when-do-seasons-start

Navaneelan, T. (2017). Suicide rates: An overview. *Statistics Canada, 2012001*(11696). https://www150.statcan.gc.ca/n1/pub/82-624-x/2012001/article/11696-eng.htm

Nemani, K., Li, C., Olfson, M., Blessing, E. M., Razavian, N., Chen, J., Petkova, E., & Goff, D. C. (2021). Association of psychiatric disorders with mortality among patients with COVID-19. *JAMA Psychiatry, 78*(4), 380–386. https://doi.org/10.1001/jamapsychiatry.2020.4442

Nielsen, S. F., Hjorthøj, C. R., Erlangsen, A., & Nordentoft, M. (2011). Psychiatric disorders and mortality among people in homeless shelters in Denmark: A nationwide register-based cohort study. *The Lancet, 377*(9784), 2205–2214. https://doi.org/10.1016/S0140-6736(11)60747-2

Nilsson, L., Tomson, T., Farahmand, B. Y., Diwan, V., & Persson, P. G. (1997). Cause-specific mortality in epilepsy: A cohort study of more than 9,000 patients once hospitalized for epilepsy. *Epilepsia, 38*(10), 1062–1068. https://doi.org/10.1111/j.1528-1157.1997.tb01194.x

Nilsson, N. H., Bendix, M., Öhlund, L., Widerström, M., Werneke, U., & Maripuu, M. (2021). Increased risks of death and hospitalization in influenza/pneumonia and sepsis for individuals affected by psychotic disorders, bipolar disorders, and single manic episodes: A retrospective cross-sectional study of Swedish national cohorts. *Journal of Clinical Medicine, 10*(19), 4411. https://doi.org/10.3390/jcm10194411

Nilsson, S. F., Hjorthøj, C. R., Erlangsen, A., & Nordentoft, M. (2014). Suicide and unintentional injury mortality among homeless people: A Danish nationwide register-based cohort study. *European Journal of Public Health, 24*(1), 50–56. https://doi.org/10.1093/eurpub/ckt025

Nishimi, K., Neylan, T. C., Bertenthal, D., et al. (2022). PTSD and risk for hospitalization and death following COVID-19 infection. *Translational Psychiatry, 12*, 482. https://doi.org/10.1038/s41398-022-02156-w

Office of the Prime Minister of Canada. (2020). *Prime Minister announces new actions under Canada's COVID-19 response*. Government of Canada.

https://www.pm.gc.ca/en/news/news-releases/2020/03/16/prime-minister-announcesnew-actions-under-canadas-covid-19-response

OpenAI. (2024). ChatGPT (GPT-4) [Large language model]. https://www.openai.com/chatgpt

Østergaard, S. D., Momen, N. C., Heide-Jørgensen, U., & Plana-Ripoll, O. (2024). Risk of suicide across medical conditions and the role of prior mental disorder. J*AMA Psychiatry, 81*(12), 1198–1206. https://doi.org/10.1001/jamapsychiatry.2024.2561

Pirkis, J., Bantjes, J., Dandona, R., Knipe, D., Pitman, A., Robinson, J., Silverman, M., & Hawton, K. (2024). Addressing key risk factors for suicide at a societal level. *Lancet Public Health, 9*(10), e816–e824. https://doi.org/10.1016/S2468-2667(24)00158-0

Pirkis, J., Gunnell, D., Shin, S., Del Pozo-Banos, M., Arya, V., Aguilar, P. A., Appleby, L., Arafat, S. M. Y., Arensman, E., Ayuso-Mateos, J. L., Balhara, Y. P. S., Bantjes, J., Baran, A., Behera, C., Bertolote, J., Borges, G., Bray, M., Brečić, P., Caine, E., ... Spittal, M. J. (2022). Suicide numbers during the first 9–15 months of the COVID-19 pandemic compared with pre-existing trends: An interrupted time series analysis in 33 countries. *EClinicalMedicine*, 51, Article 101573. https://doi.org/10.1016/j.eclinm.2022.101573

Pospisil, L. J. (1958). *Kapauku Papuans and their law*. Published for the Dept. Anthropology, Yale University, New Haven.

Potter, A. L., Haridas, C., Neumann, K., Kiang, M. V., Fong, Z. V., Riddell, C. A., … Yang, C.-F. J. (2023). Incidence, timing, and factors associated with suicide among patients undergoing surgery for cancer in the US. *JAMA Oncology, 9*(3), 308–315. https://doi.org/10.1001/jamaoncol.2022.6549

Qin, P., Waltoft, B. L., Mortensen, P. B., & Postolache, T. T. (2013). Suicide risk in relation to air pollen counts: A study based on data from Danish registers. *BMJ Open, 3*(5), e002462. https://doi.org/10.1136/bmjopen-2012-002462

R Core Team. (2023). *R: A language and environment for statistical computing.* R Foundation for Statistical Computing. https://www.r-project.org/

Romero Starke, K., Reissig, D., Petereit-Haack, G., Schmauder, S., Nienhaus, A., & Seidler, A. (2021). The isolated effect of age on the risk of COVID-19 severe outcomes: A systematic review with meta-analysis. *BMJ Global Health, 6*(12). https://doi.org/10.1136/bmjgh-2021-006434

Ronaldson, A., Nascimento Santana, I., Carlisle, S., Atmore, K. H., Chilman, N., Heslin, M., Markham, S., Dregan, A., Das-Munshi, J., Lampejo, T., Hotopf, M., & Bakolis, I. (2024). Severe mental illness and infectious disease mortality: A systematic review and meta-analysis. *EClinicalMedicine, 77*, 102867. https://doi.org/10.1016/j.eclinm.2024.102867

Raudenbush, S. W., & Bryk, A. S. (2002). *Hierarchical linear models: Applications and data analysis methods* (2nd ed.). SAGE Publications.

Raschka, S., & Mirjalili, V. (2017). *Python machine learning.* Packt Publishing.

Roerecke, M., & Rehm, J. (2013). Alcohol use disorders and mortality: A systematic review and meta-analysis. *Addiction, 108*(9), 1562–1578. https://doi.org/10.1111/add.12231

Rosenthal, R. W. (1993). Suicide attempts and signalling games. *Mathematical Social Sciences, 26,* 25–33. https://doi.org/10.1016/0165-4896(93)90009-8

Schanzer, D. L., Sevenhuysen, C., Winchester, B., & Mersereau, T. (2013). Estimating influenza deaths in Canada, 1992–2009. *PLOS ONE, 8*(11), Article e80481. https://doi.org/10.1371/journal.pone.0080481

Shapiro, M. (2019). Suicide rates spike in spring, not winter. Johns Hopkins Medicine. https://www.hopkinsmedicine.org/news/articles/2019/05/suicide-rates-spike-in-spring-not-winter

Sheth, T., Nair, C., Muller, J., & Yusuf, S. (1999). Increased winter mortality from acute myocardial infarction and stroke: The effect of age. *Journal of the American College of Cardiology*, *33*(7), 1916–1919. https://doi.org/10.1016/S0735-1097(99)00137-0

Siahaan, Y. M. T., Ketaren, R. J., Hartoyo, V., & Hariyanto, T. I. (2021). Epilepsy and the risk of severe COVID-19 outcomes: Systematic review and meta-analysis. *Epilepsy & Behavior, 125*, 108437. https://doi.org/10.1016/j.yebeh.2021.108437

Sinyor, M., Knipe, D., Borges, G., Ueda, M., Pirkis, J., Phillips, M. R., Gunnell, D., & International COVID-19 Suicide Prevention Research Collaboration (2022). Suicide risk and prevention during the COVID-19 pandemic: one year on. *Archives of Suicide Research, 26*(4), 1944-1949. https://doi.org/10.1080/13811118.2021.1955784

Smith, P. N., & Cukrowicz, K. C. (2010). Capable of suicide: A functional model of the acquired capability component of the Interpersonal-Psychological Theory of Suicide. *Suicide and Life-Threatening Behavior, 40*(3), 266–275. https://onlinelibrary.wiley.com/doi/abs/10.1521/suli.2010.40.3.266

Soper, C. A. (2018). *The evolution of suicide*. Springer.

Soper, C. A., & Shackelford, T. K. (2023). Are depression and suicidality evolved signals? Evidently, no. *Evolution and Human Behavior, 44*(2), 147–154. https://doi.org/10.1016/j.evolhumbehav.2023.02.004

Spence, M. (1973). Job market signaling. *The Quarterly Journal of Economics, 87*(3), 355–374. https://doi.org/10.2307/1882010

Statistics Canada. (2021). *Provisional death counts and excess mortality, January 2020 to May 2021*. The Daily. https://www150.statcan.gc.ca/n1/daily-quotidien/210809/dq210809a-eng.htm

Statistics Canada. (2026a). Provisional weekly death counts, by selected grouped causes of death (Table 13-10-0810-01). https://doi.org/10.25318/1310081001-eng

Statistics Canada. (2026b). Population estimates, quarterly (Table 17-10-0009-01). https://doi.org/10.25318/1710000901-eng

Statistics Canada. (2026c). Quarterly demographic estimates (QDE). Retrieved from https://www23.statcan.gc.ca/imdb/p2SV.pl?Function=getSurvey&SDDS=3601

Stengel, E. (1952). Enquiries into attempted suicide [Abridged]. *Proceedings of the Royal Society of Medicine, 45,* 613–620. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1987521/

Stengel, E. (1964). *Suicide and attempted suicide.* Penguin Books.

Syme, K. L., Garfield, Z. H., & Hagen, E. H. (2016). Testing the bargaining vs. inclusive fitness models of suicidal behavior against the ethnographic record. *Evolution and Human Behavior, 37*(3), 179–192. https://doi.org/10.1016/j.evolhumbehav.2015.10.005

Tan, Z. Q. (2021). Testing evolutionary theories of suicidal ideation: Bargaining, costly apology, and inclusive fitness (Master's thesis). https://zenodo.org/records/5039496

The Canadian Press. (2020, March 2). Timeline of COVID-19 cases across Canada. *CBC News*. Retrieved June 11, 2025, from https://www.cbc.ca/news/health/canada-coronavirus-timeline-1.5482310

Tsiatis, A. A. (1975). A nonidentifiability aspect of the problem of competing risks. *Proceedings of the National Academy of Sciences of the United States of America, 72*(1), 20–22. https://doi.org/10.1073/pnas.72.1.20

Vai, B., Mazza, M. G., Delli Colli, C., Foiselle, M., Allen, B., Benedetti, F., … De Picker, L. (2021). Mental disorders and risk of COVID-19-related mortality, hospitalisation, and intensive care unit admission: A systematic review and meta-analysis. *The Lancet Psychiatry, 8*(9), 797–812. https://doi.org/10.1016/S2215-0366(21)00232-7

Vaupel, J. W., Manton, K. G., & Stallard, E. (1979). The impact of heterogeneity in individual frailty on the dynamics of mortality. *Demography, 16*(3), 439–454. https://doi.org/10.2307/2061224

Venkatesulu, B. P., Chandrasekar, V. T., Girdhar, P., et al. (2021). A systematic review and meta-analysis of cancer patients affected by a novel coronavirus. *JNCI Cancer Spectrum, 5*(2), pkaa102. https://doi.org/10.1093/jncics/pkaa102

Walker, E. R., McGee, R. E., & Druss, B. G. (2015). Mortality in mental disorders and global disease burden: A systematic review and meta-analysis. *JAMA Psychiatry, 72*(4), 334–341. https://doi.org/10.1001/jamapsychiatry.2014.2502

Wang, H., Chen, Y., Chen, S., & Xiao, Z. (2023). Suicidality and epilepsy: A systematic review and meta-analysis. *Frontiers in Psychiatry, 14*, 1097516. https://doi.org/10.3389/fpsyt.2023.1097516

Wang, Q. Q., Kaelber, D. C., Xu, R., & Volkow, N. D. (2021). COVID-19 risk and outcomes in patients with substance use disorders: Analyses from electronic health records in the United States. *Molecular Psychiatry, 26*(1), 30–39. https://doi.org/10.1038/s41380-020-00880-7

WeatherSpark. (2025). *Average weather at Ottawa Macdonald-Cartier International Airport, Ontario, Canada year round.* WeatherSpark. Retrieved May 27, 2025, from https://weatherspark.com/y/147151/Average-Weather-at-Ottawa-Macdonald-Cartier-International-Airport-Ontario-Canada-Year-Round#Figures-Temperature

Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis* (Version 3.4.0) [Computer software]. Springer. https://ggplot2.tidyverse.org

Wickham, H., François, R., Henry, L., & Müller, K. (2023). dplyr: A grammar of data manipulation (Version 1.1.3) [Computer software]. https://dplyr.tidyverse.org

Wiley, J. C. (2020). Psychological aposematism: An evolutionary analysis of suicide. *Biological Theory, 15*(3), 226–238. https://doi.org/10.1007/s13752-020-00353-8

Woo, J.-M., Okusaga, O., & Postolache, T. T. (2012). Seasonality of suicidal behavior. *International Journal of Environmental Research and Public Health, 9*(2), 531–547. https://doi.org/10.3390/ijerph9020531

World Health Organization. (2023). Coronavirus disease (COVID-19) situation reports. *World Health Organization*. Retrieved December 14, 2024, from https://www.who.int/emergencies/diseases/novel-coronavirus-2019/situation-reports

Xu, Y.-B., Liu, H., Cao, Q.-H., Ji, J.-L., Dong, R.-R., & Xu, D. (2020). Evaluating overall survival and competing risks of survival in patients with early-stage breast cancer using a comprehensive nomogram. *Cancer Medicine, 9*(12), 4095–4106. https://doi.org/10.1002/cam4.3030

Yu, B., & Chen, X. (2019). Age and birth cohort–adjusted rates of suicide mortality among US male and female youths aged 10 to 19 years from 1999 to 2017. *JAMA Network Open, 2*(9), e1911383. https://doi.org/10.1001/jamanetworkopen.2019.11383

Zalcman, R. F., & Mann, R. E. (2007). The effects of privatization of alcohol sales in Alberta on suicide mortality rates. *Contemporary Drug Problems, 34*(4), 589–609. https://doi.org/10.1177/009145090703400405

Zeileis, A., & Grothendieck, G. (2005). *zoo: S3 infrastructure for regular and irregular time series*. R package version 1.8-12. https://CRAN.R-project.org/package=zoo

Zhong, S., Senior, M., Yu, R., Perry, A., Hawton, K., Shaw, J., & Fazel, S. (2021). Risk factors for suicide in prisons: An updated systematic review and meta-analysis. *The Lancet Public Health, 6*(3), e164–e174. https://doi.org/10.1016/S2468-2667(20)30233-4

Zlodre, J., & Fazel, S. (2012). All-cause and external mortality in released prisoners: Systematic review and meta-analysis. *American Journal of Public Health, 102*(12), e67–e75. https://doi.org/10.2105/AJPH.2012.300764

Figure 1. Weekly time series plots of Standardised suicide rates and death rates by year

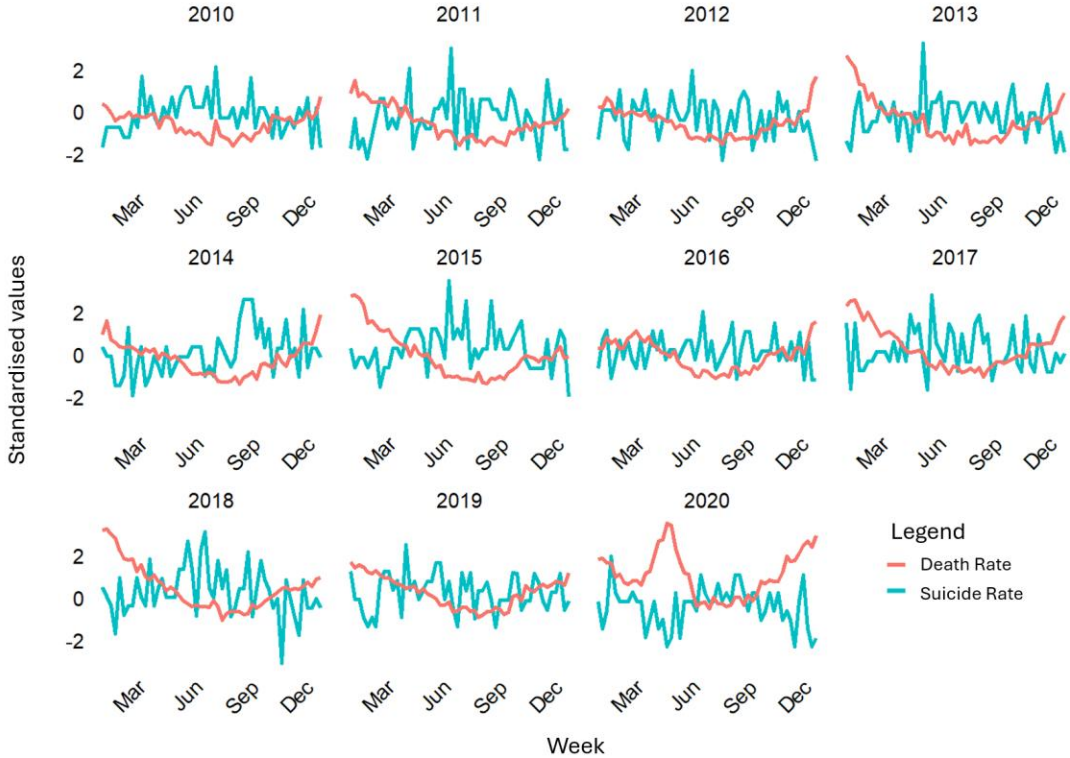


Figure 2. Weekly suicide rates and death rates averaged by month, Standardised

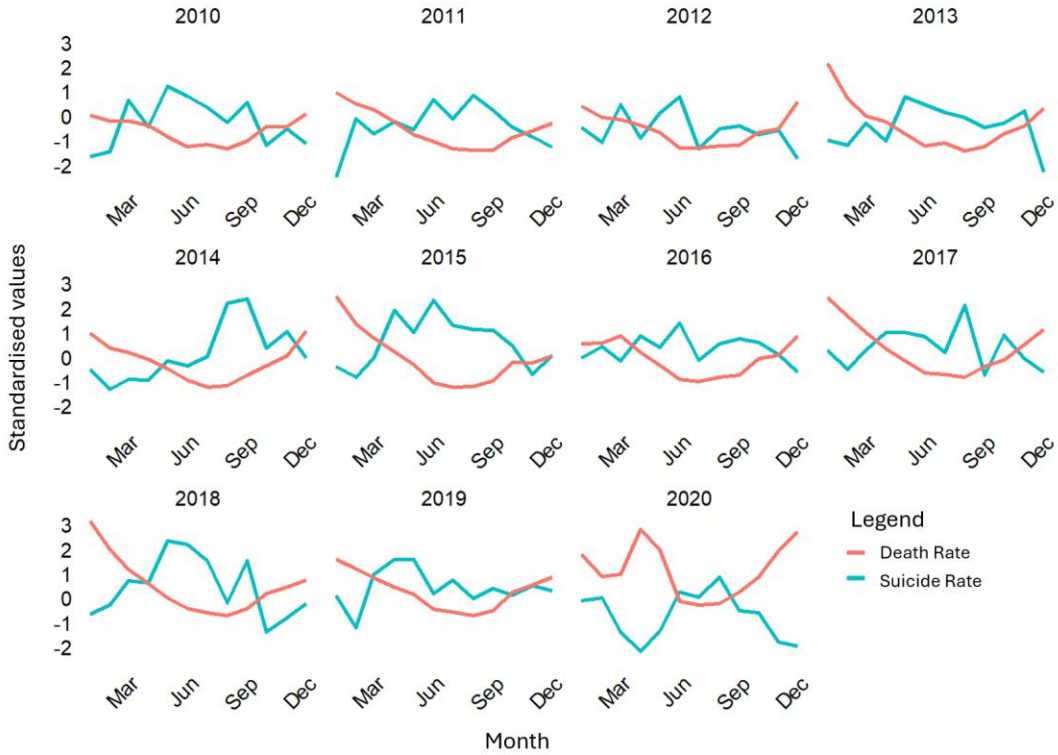
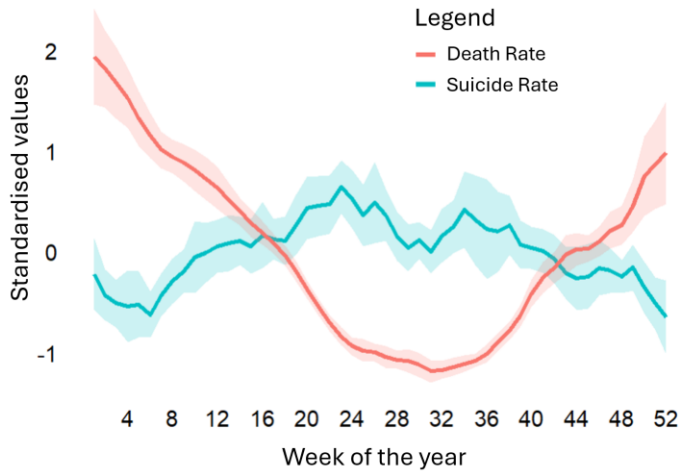
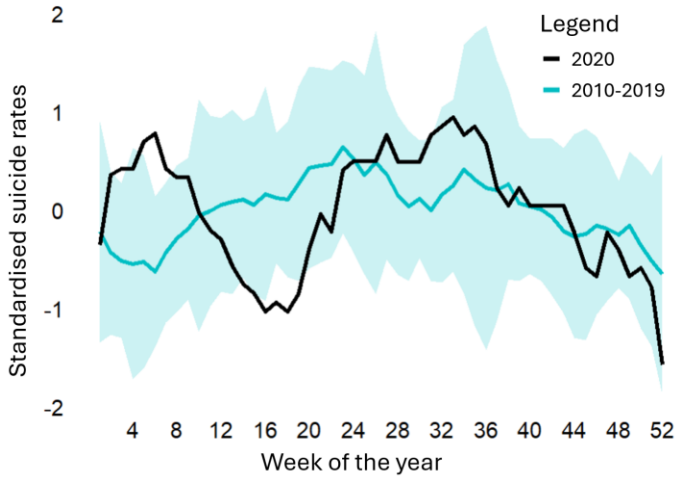


Figure 3. Five-week moving averages for death and suicide rates, 2010 to 2019



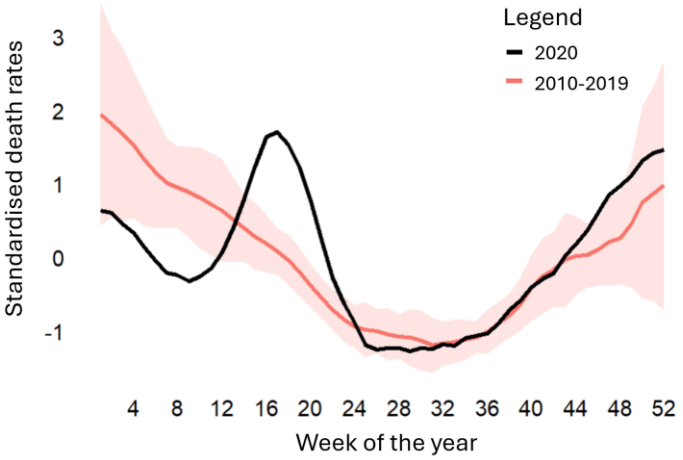
Note. Lighter shading indicates 95% confidence bands.

Figure 4. Five-week moving averages for suicide rates, comparing 2020 to historical seasonality



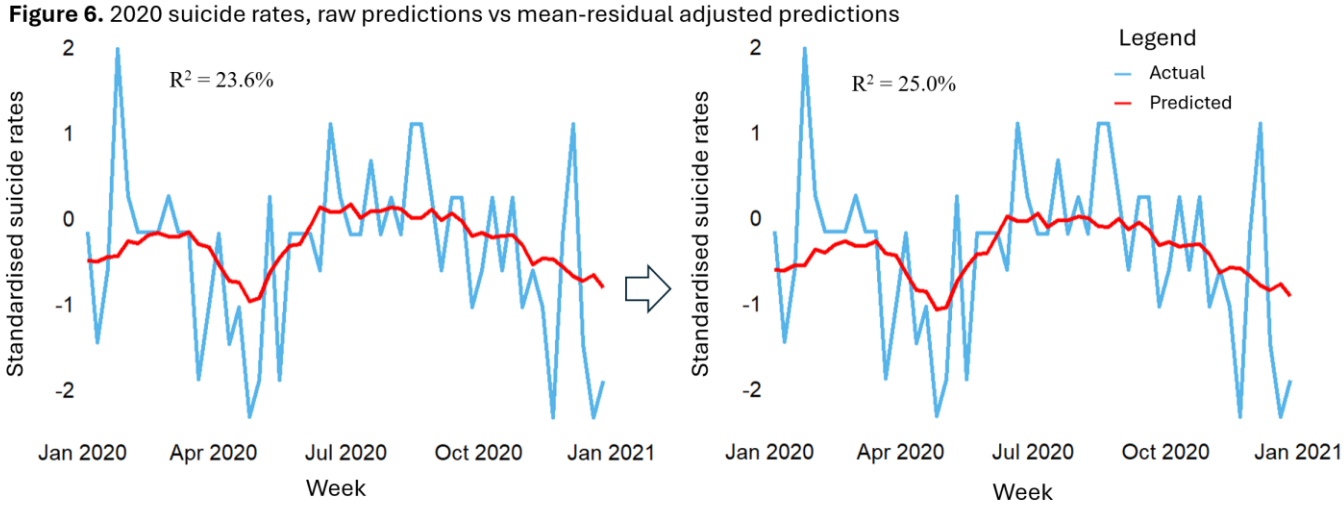
Note. Lighter shading indicates a 95% prediction band.

Figure 5. Five-week moving averages for death rates, comparing 2020 to historical seasonality



Note. Lighter shading indicates a 95% prediction band.

Figure 6. 2020 suicide rates, raw predictions vs mean-residual adjusted predictions



Note. The time series plot displayed on the left showcases the raw predicted values from our model, while the time series plot displayed on the right showcases the mean-residual adjusted predictions, for 2020 data.